

Data Wrangling

Complete Process

The E FD VO IN Pipeline E - Encoding (Sex, Embarked) --- 3 F - Feature Engineering (Creating Family Size) D - Deduplication (Removing duplicates) V - Visualization (Plotting Box plots to inspect distribution) O - Outlier Handling (Capping Fare) I - Imputation (Filling missing Age) N - Normalization (StandardScaler)

```
In [3]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler

# -----
# LOAD DATA
# -----
titanic = sns.load_dataset('titanic')
#

# -----
# STEP 1: HANDLE CATEGORICAL VARIABLES
# -----
print("Processing Categorical Variables...")
label_encoder = LabelEncoder()

# Encode 'sex'
titanic['sex'] = label_encoder.fit_transform(titanic['sex'])

# Encode 'embarked' (filling NaNs with mode 'S' first to prevent errors)
titanic['embarked'] = titanic['embarked'].fillna('S')
titanic['embarked'] = label_encoder.fit_transform(titanic['embarked'])

# Convert 'who' into binary (man: 1, woman: 0)
titanic['who'] = titanic['who'].apply(lambda x: 1 if x == 'man' else 0)

# -----
# STEP 2: FEATURE ENGINEERING
# -----
print("Creating New Features...")
titanic['family_size'] = titanic['sibsp'] + titanic['parch'] + 1

# -----
# STEP 3: REMOVE DUPLICATES
# -----
print("Removing Duplicates...")
titanic_cleaned = titanic.drop_duplicates().copy()

# -----
# STEP 4: HANDLE OUTLIERS
```

```

# -----
print("Handling Outliers...")

# --- VISUALIZATION RESTORED HERE ---
plt.figure(figsize=(8, 4))
plt.boxplot(titanic_cleaned['fare'])
plt.title("Box Plot for Fare (Outlier Detection)")
plt.xlabel("Fare")
plt.show() # This will open a window with the plot

# Cap outliers in 'fare' to the 99th percentile
fare_cap = titanic_cleaned['fare'].quantile(0.99)
titanic_cleaned['fare'] = np.where(
    titanic_cleaned['fare'] > fare_cap,
    fare_cap,
    titanic_cleaned['fare']
)

print("\nFare column statistics after handling outliers:")
print(titanic_cleaned['fare'].describe())

# -----
# STEP 5: HANDLE MISSING VALUES
# -----
print("Imputing Missing Values...")
titanic_cleaned['age'] = titanic_cleaned['age'].fillna(titanic_cleaned['age'].median())

# -----
# STEP 6: NORMALIZE NUMERICAL DATA
# -----
print("Normalizing Data...")
scaler = StandardScaler()
titanic_cleaned[['age', 'fare']] = scaler.fit_transform(titanic_cleaned[['age', 'fare']])

# -----
# STEP 7: SAVE DATA
# -----
titanic_cleaned.to_csv('titanic_cleaned.csv', index=False)
print("\nCleaned dataset saved to 'titanic_cleaned.csv'")
print(titanic_cleaned.head())

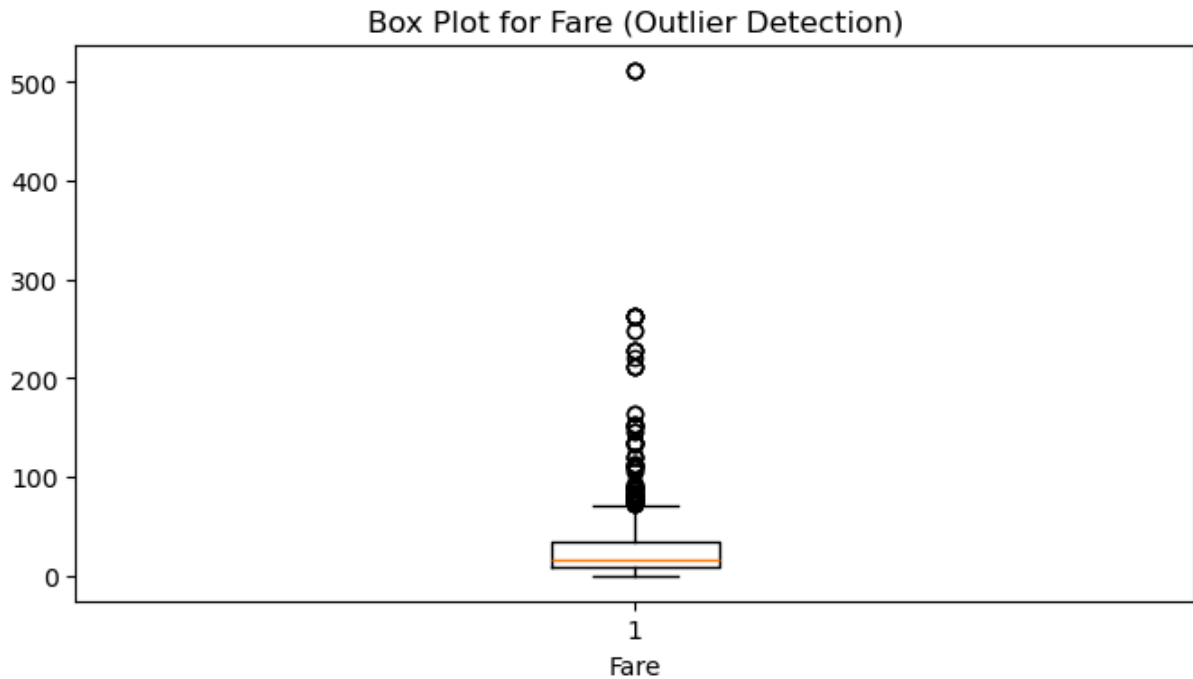
```

Processing Categorical Variables...

Creating New Features...

Removing Duplicates...

Handling Outliers...



Fare column statistics after handling outliers:

```
count    784.000000
mean      33.752093
std       45.198131
min        0.000000
25%       8.050000
50%      15.900000
75%      34.109350
max      262.375000
```

Name: fare, dtype: float64

Imputing Missing Values...

Normalizing Data...

Cleaned dataset saved to 'titanic_cleaned.csv'

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class \
0	0	3	1	-0.557360	1	0	-0.586728	2	Third
1	1	1	0	0.608298	1	0	0.830901	0	First
2	1	3	0	-0.265945	0	0	-0.571784	2	Third
3	1	1	0	0.389737	1	0	0.428342	2	First
4	0	3	1	0.389737	0	0	-0.569017	2	Third

	who	adult_male	deck	embark_town	alive	alone	family_size
0	0	True	NaN	Southampton	no	False	2
1	0	False	C	Cherbourg	yes	False	2
2	0	False	NaN	Southampton	yes	True	1
3	0	False	C	Southampton	yes	False	2
4	0	True	NaN	Southampton	no	True	1

How to remember code each step is hint for following step

In []: `# I LE FD VO INS - Complete Data Wrangling`

```
In [5]: # Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load the Dataset
titanic = sns.load_dataset("titanic")
titanic.head()
```

```
Out[5]:
```

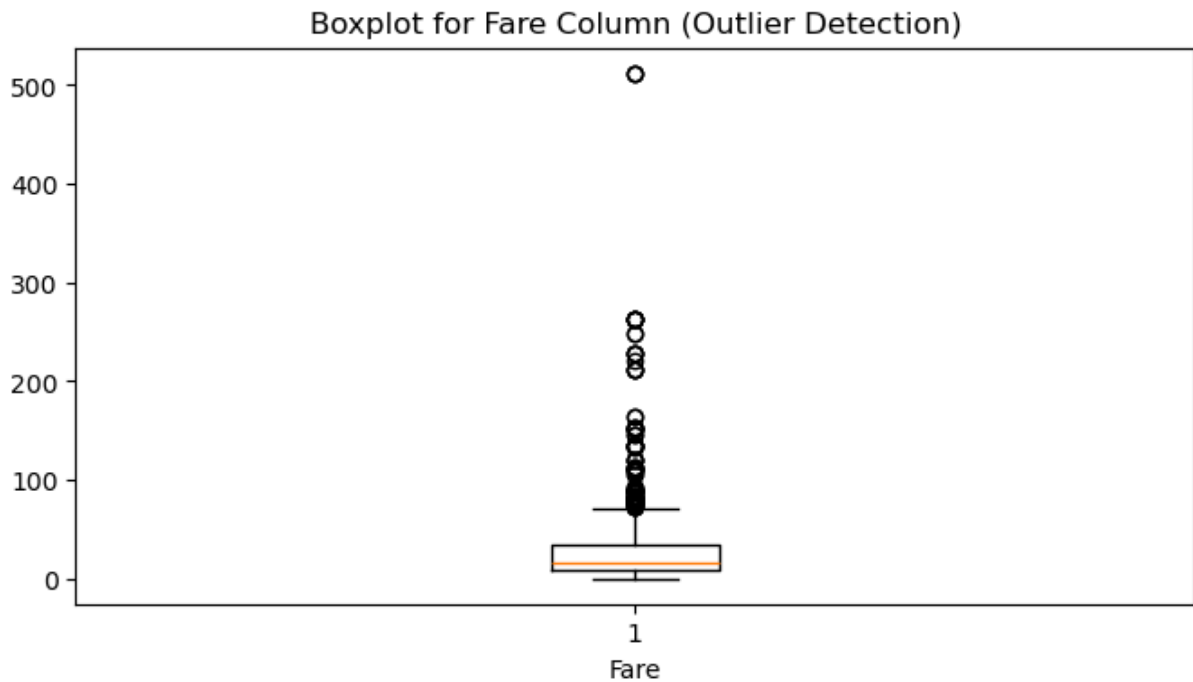
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

```
In [10]: # Encoding - for the categorical columns like sex, embarked, who in above look ↑
label_encoder = LabelEncoder()
titanic['sex'] = label_encoder.fit_transform(titanic['sex'])
titanic['embarked'] = titanic['embarked'].fillna('S')
titanic['embarked'] = label_encoder.fit_transform(titanic['embarked'])
titanic['who'] = titanic['who'].apply(lambda x: 1 if x == "man" else 0)
```

```
In [11]: # Feature Engineering - make family column
titanic['family'] = titanic['sibsp'] + titanic['parch'] + 1
```

```
In [12]: # De-Duplication - remove duplicates values and make copy to keep the titanic varia
titanic_cleaned = titanic.drop_duplicates().copy()
```

```
In [15]: # Visualization
plt.figure(figsize=(8,4))
plt.boxplot(titanic_cleaned['fare'])
plt.title("Boxplot for Fare Column (Outlier Detection)")
plt.xlabel("Fare")
plt.show()
```



```
In [17]: # Outlier Detection - set a standard | store the Logic in variable | show outliers
fare_cap = titanic_cleaned['fare'].quantile(0.99) # standard
titanic_cleaned['fare'] = np.where(titanic_cleaned['fare'] > fare_cap, fare_cap, titanic_cleaned['fare'])
print(titanic_cleaned['fare'].describe()) # show outliers
```

```
count    784.000000
mean      33.752093
std       45.198131
min        0.000000
25%        8.050000
50%       15.900000
75%       34.109350
max      262.375000
Name: fare, dtype: float64
```

```
In [18]: # Imputation
titanic_cleaned['age'] = titanic_cleaned['age'].fillna(titanic_cleaned['age'].median())
```

```
In [20]: # Normalization - rescaling the specific columns so that the biased data does not dominate
scaler = StandardScaler()
titanic_cleaned[['age', 'fare']] = scaler.fit_transform(titanic_cleaned[['age', 'fare']])
```

```
In [21]: # save to csv file
csv_file_path = "titanic_cleaned.csv"
titanic_cleaned.to_csv(csv_file_path, index=False)
print(f"The Cleaned Data is saved at {csv_file_path}")
```

The Cleaned Data is saved at titanic_cleaned.csv

```
In [24]: # Full Code - I LE FD VO INS - steps
# Import
import pandas as pd
import seaborn as sns
import numpy as np
```

```

import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load
titanic = sns.load_dataset('titanic')
titanic.head() # for reference lookup for each step

# Encoding - 3
label_encoder = LabelEncoder()
titanic['sex'] = label_encoder.fit_transform(titanic['sex'])
titanic['embarked'] = titanic['embarked'].fillna('S')
titanic['embarked'] = label_encoder.fit_transform(titanic['embarked'])
titanic['who'] = titanic['who'].apply(lambda x: 1 if x == 'man' else 0)

# Feature Engineering - family from sibsp + parch
titanic['family'] = titanic['sibsp'] + titanic['parch'] + 1

# De-Duplication - remove duplicates on copy
titanic_cleaned = titanic.drop_duplicates().copy()

# Visualization
plt.figure(figsize=(8,4))
plt.boxplot(titanic_cleaned['fare'])
plt.title("Boxplot for Fare (Outlier Detection)")
plt.xlabel("Fare")
plt.show()

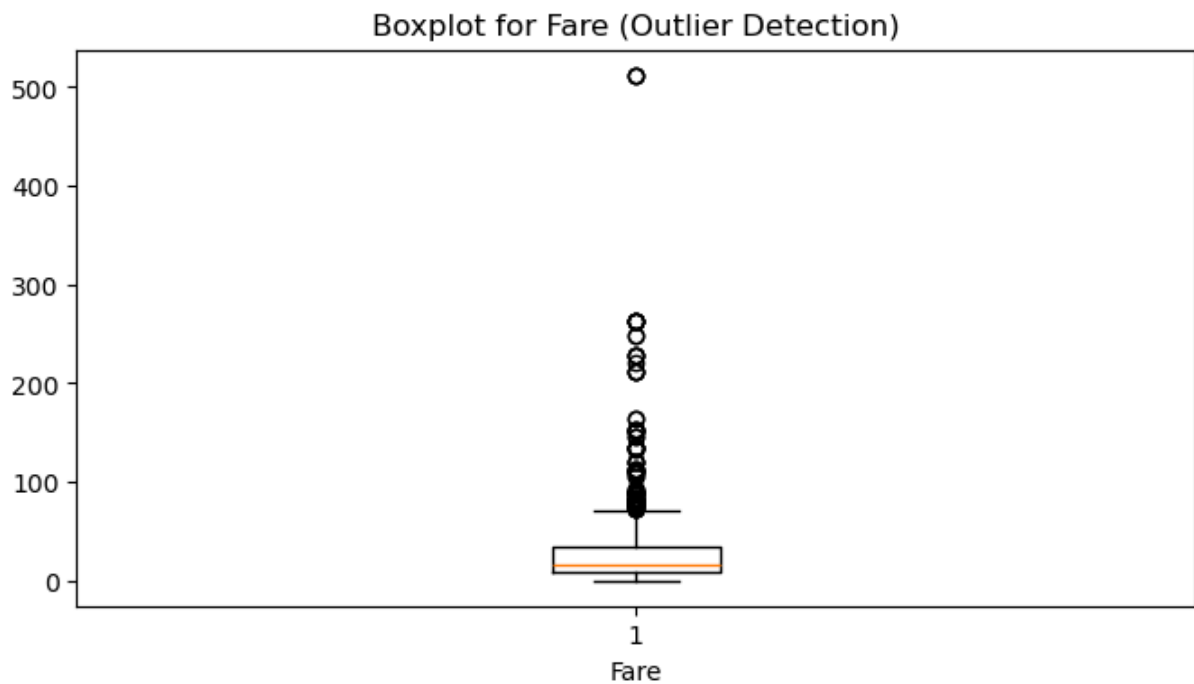
# Outlier Detection
fare_cap = titanic_cleaned['fare'].quantile(0.99)
titanic_cleaned['fare'] = np.where(titanic_cleaned['fare'] > fare_cap, fare_cap, titanic_cleaned['fare'])
print(titanic_cleaned['fare'].describe())

# Imputation
titanic_cleaned['age'] = titanic_cleaned['age'].fillna(titanic_cleaned['age'].median())

# Normalization
scaler = StandardScaler()
titanic_cleaned[['age', 'fare']] = scaler.fit_transform(titanic_cleaned[['age', 'fare']])

# Save data to CSV
csv_file_path = "titanic_cleaned.csv"
titanic_cleaned.to_csv(csv_file_path, index=False)
print(f"CSV file saved at:{csv_file_path}")
print(titanic_cleaned.head())

```



```
count    784.000000
mean      33.752093
std       45.198131
min        0.000000
25%        8.050000
50%       15.900000
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max      262.375000
```

Name: fare, dtype: float64

CSV file saved at:titanic_cleaned.csv

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4	1	True	NaN	Southampton	no	True	1

In []: